

8-2

1.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.



Using correct units, interpret the meaning of $\int_0^{2.4} v(t) dt$ in the context of the problem. Approximate

$\int_0^{2.4} v(t) dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.

Part B

The response can earn up to 3 points:

1 point: For the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth, traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point: For use of the midpoint Riemann sum

1 point: For the correct approximation

$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) = 36.64$ miles



0	1	2	3
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The response earns all three of the following points:

1 point: For the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth, traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point: For use of the midpoint Riemann sum

1 point: For the correct approximation

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$$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) = 36.64 \text{ miles}$$

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2. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

A snail is traveling along a straight path. The snail's velocity can be modeled by $v(t) = 1.4 \ln(1 + t^2)$ inches per minute for $0 \leq t \leq 15$ minutes.



- (a) Find the acceleration of the snail at time $t = 5$ minutes.
- (b) What is the displacement of the snail over the interval $0 \leq t \leq 15$ minutes?
- (c) At what time t , $0 \leq t \leq 15$, is the snail's instantaneous velocity equal to its average velocity over the interval $0 \leq t \leq 15$?
- (d) An ant arrives at the snail's starting position at time $t = 12$ minutes and follows the snail's path. During the interval $12 \leq t \leq 15$ minutes, the ant travels in the same direction as the snail with a constant acceleration of 2 inches per minute per minute. The ant catches up to the snail at time $t = 15$ minutes. The ant's velocity at time $t = 12$ is B inches per minute. Find the value of B .

Part A

1 point is earned for: answer

$$v'(5) = 0.538462$$

The acceleration of the snail at time $t = 5$ minutes is 0.538 inches per minute per minute.



0	1
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The student response earns all of the following points:

1 point is earned for: answer

$$v'(5) = 0.538462$$

The acceleration of the snail at time $t = 5$ minutes is 0.538 inches per minute per minute.

Part B

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{15} v(t) dt = 76.043074$$

The displacement of the snail over the interval $0 \leq t \leq 15$ minutes is 76.043 inches.

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0	1	2
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The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{15} v(t) \, dt = 76.043074$$

The displacement of the snail over the interval $0 \leq t \leq 15$ minutes is 76.043 inches.

Part C

1 point is earned for: average value expression

1 point is earned for: answer

$$\frac{1}{15} \int_0^{15} v(t) \, dt = 5.069538$$

$$1.4 \ln(1 + t^2) = 5.069538 \Rightarrow t = 6.031 \text{ minutes}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: average value expression

1 point is earned for: answer

$$\frac{1}{15} \int_0^{15} v(t) \, dt = 5.069538$$

$$1.4 \ln(1 + t^2) = 5.069538 \Rightarrow t = 6.031 \text{ minutes}$$

Part D

1 point is earned for: ant's velocity

1 point is earned for: ant's displacement

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1 point is earned for: equation**1 point is earned for:** answer

The velocity of the ant at time t , $12 \leq t \leq 15$, is $\int 2 \, dt = 2t + c$ inches per minute for some constant c .

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2t + c) \, dt = (t^2 + ct) \Big|_{t=12}^{t=15} = 81 + 3c$ inches.

Thus, $81 + 3c = 76.043074 \Rightarrow c = -1.652309$.

The velocity of the ant at time $t = 12$ is $B = 2 \cdot 12 - 1.652309 = 22.348$ (or 22.347) inches per minute.

— OR —

The velocity of the ant at time t , $12 \leq t \leq 15$, is $2(t - 12) + B$ inches per minute.

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2(t - 12) + B) \, dt = ((t - 12)^2 + Bt) \Big|_{t=12}^{t=15} = 9 + 3B$ inches.

$9 + 3B = 76.043074 \Rightarrow B = 22.348$ (or 22.347) inches per minute



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: ant's velocity**1 point is earned for:** ant's displacement**1 point is earned for:** equation**1 point is earned for:** answer

The velocity of the ant at time t , $12 \leq t \leq 15$, is $\int 2 \, dt = 2t + c$ inches per minute for some constant c .

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2t + c) \, dt = (t^2 + ct) \Big|_{t=12}^{t=15} = 81 + 3c$ inches.

Thus, $81 + 3c = 76.043074 \Rightarrow c = -1.652309$.

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The velocity of the ant at time $t = 12$ is $B = 2 \cdot 12 - 1.652309 = 22.348$ (or 22.347) inches per minute.

— OR —

The velocity of the ant at time t , $12 \leq t \leq 15$, is $2(t - 12) + B$ inches per minute.

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2(t - 12) + B) dt = \left((t - 12)^2 + Bt \right) \bigg|_{t=12}^{t=15} = 9 + 3B$ inches.

$$9 + 3B = 76.043074 \Rightarrow B = 22.348 \text{ (or 22.347) inches per minute}$$

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3. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

A particle moves along the x -axis so that its velocity at time t is given by $v(t) = \frac{t^6 - 13t^4 + 12}{10t^3 + 3}$. At time $t = 0$, the initial position of the particle is $x = 7$.



(a) Find the acceleration of the particle at time $t = 5.1$.

(b) Find all values of t in the interval $0 \leq t \leq 2$ for which the speed of the particle is 1.

(c) Find the position of the particle at time $t = 4$. Is the particle moving toward the origin or away from the origin at time $t = 4$? Justify your answer.

(d) During the time interval $0 < t \leq 4$, does the particle return to its initial position? Give a reason for your answer.

Part A

1 point is earned for: answer

$$v'(5.1) = 6.491816$$

The acceleration of the particle at time $t = 5.1$ is 6.492 (or 6.491).



0	1
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The student response earns all of the following points:

1 point is earned for: answer

$$v'(5.1) = 6.491816$$

The acceleration of the particle at time $t = 5.1$ is 6.492 (or 6.491).

Part B

1 point is earned for: considers $|v(t)| = 1$

1 point is earned for: answer

$$|v(t)| = 1 \text{ for } 0 \leq t \leq 2 \Rightarrow t = 0.771829, t = 1.400556$$

The speed of the particle is 1 at times $t = 0.772$ (or 0.771) and $t = 1.401$ (or 1.400).



0	1	2
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The student response earns all of the following points:

1 point is earned for: considers $|v(t)| = 1$

1 point is earned for: answer

$$|v(t)| = 1 \text{ for } 0 \leq t \leq 2 \Rightarrow t = 0.771829, t = 1.400556$$

The speed of the particle is 1 at times $t = 0.772$ (or 0.771) and $t = 1.401$ (or 1.400).

Part C

1 point is earned for: integral

1 point is earned for: uses initial condition

1 point is earned for: position

1 point is earned for: movement relative to the origin, with justification

Let $x(t)$ be the position of the particle at time t .

$$x(4) = 7 + \int_0^4 v(t) dt = 6.711558$$

The position of the particle at time $t = 4$ is 6.712 (or 6.711).

The particle is moving away from the origin because $x(4) > 0$ and $v(4) = 1.213064 > 0$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: uses initial condition

1 point is earned for: position

1 point is earned for: movement relative to the origin, with justification

Let $x(t)$ be the position of the particle at time t .

$$x(4) = 7 + \int_0^4 v(t) dt = 6.711558$$

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The position of the particle at time $t = 4$ is 6.712 (or 6.711).

The particle is moving away from the origin because $x(4) > 0$ and $v(4) = 1.213064 > 0$.

part D

1 point is earned for: sandwiches initial position

1 point is earned for: answer with reason

$x'(t) = v(t) \Rightarrow x(t)$ is continuous on $0 < t \leq 4$.

$$x(1) = 7 + \int_0^1 v(t) dt = 9.403593$$

From part (c), $x(4) = 6.712$.

From time $t = 1$ to $t = 4$, the particle moves from $x = 9.404$ to $x = 6.712$. By the Intermediate Value Theorem, the particle must return to $x = 7$ during the time interval.



0	1	2
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The student response earns all of the following points:

1 point is earned for: sandwiches initial position

1 point is earned for: answer with reason

$x'(t) = v(t) \Rightarrow x(t)$ is continuous on $0 < t \leq 4$.

$$x(1) = 7 + \int_0^1 v(t) dt = 9.403593$$

From part (c), $x(4) = 6.712$.

From time $t = 1$ to $t = 4$, the particle moves from $x = 9.404$ to $x = 6.712$. By the Intermediate Value Theorem, the particle must return to $x = 7$ during the time interval.

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4. A particle moves along the x-axis so that at time $t \geq 0$, the acceleration of the particle is $a(t) = 15\sqrt{t}$. The position of the particle is 10 when $t = 0$, and the position of the particle is 20 when $t = 1$. What is the velocity of the particle at time $t = 0$?
- (A) -14
(B) 0
(C) 5
(D) 6 ✓
(E) 10
5. A particle moves along the x-axis so that at any time $t > 0$, its velocity is given by $v(t) = 4 - 6t^2$. If the particle is at position $x = 7$ at $t = 1$ time, what is the position of the particle at time $t = 2$?
- (A) -10
(B) -5
(C) -3 ✓
(D) 3
(E) 17
6. A particle moves along the x-axis. The velocity of the particle at time t is given by $v(t) = \frac{4}{t^3+1}$.
-  If the position of the particle is $x = 1$ when $t = 2$, what is the position of the particle when $t = 4$?
- (A) 0.617
(B) 0.647
(C) 1.353 ✓
(D) 5.713
7. A particle moves along the x-axis so that at any time t , $t \geq 0$, its acceleration is $a(t) = -4 \sin(2t)$. If the velocity of the particle at $t = 0$ is $v(0) = 7$ and its position at $t = 0$ is $x(0) = 0$, then its position at time t is $x(t) =$
- (A) $\sin(2t) + 5t$ ✓
(B) $\sin(2t) + 7t$
(C) $\sin(2t) + 9t$
(D) $16\sin(2t) + 7t$

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8. A particle moves along the x -axis so that at any time $t \geq 0$, its velocity is given by $v(t) = \sin(2t)$. If the position of the particle at time $t = \frac{\pi}{2}$ is $x = 4$, what is the particle's position at time $t = 0$?
- (A) $-\frac{1}{2}$
- (B) 2
- (C) 3
- (D) 5
- (E) 8
9.  A particle moves along the x -axis so that its velocity at time $t \geq 0$ is given by $v(t) = \frac{t^2 - 1}{t^2 + 1}$. What is the total distance traveled by the particle from $t = 0$ to $t = 2$?
- (A) 0.214
- (B) 0.320
- (C) 0.600
- (D) 0.927
- (E) 1.600

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10. A toy train moves along a straight set up on a table. The position $x(t)$ of the train at the time t seconds is measured in centimeters from the center of the track. At time $t=1$, the train is 6 centimeters to the left of the center, so $x(1)=-6$. For $0 \leq t \leq 4$, the velocity of the train at the time t is given by $v(t)=3t^2-12$, where $v(t)$ is measured in centimeters per second.

For $0 \leq t \leq 4$, find $x(t)$.

Part A

The response can earn up to 3 points:

1 point: For the correct integral

1 point: For the correct antiderivative

1 point: For the correct answer

$$x(t) = -6 + \int_1^t (3u^2 - 12) du = -6 + [u^3 - 12u]_{u=1}^{u=t} = t^3 - 12t + 5$$



0	1	2	3
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The response earns all three of the following points:

1 point: For the correct integral

1 point: For the correct antiderivative

1 point: For the correct answer

$$x(t) = -6 + \int_1^t (3u^2 - 12) du = -6 + [u^3 - 12u]_{u=1}^{u=t} = t^3 - 12t + 5$$

11. A particle moves along the x -axis with velocity given by $v(t) = 3t^2 - 4$ for time $t \geq 0$. If the particle is at position $x = -2$ at time $t = 0$, what is the position of the particle at the time $t = 3$?

(A) 13

(B) 15

(C) 16

(D) 17

(E) 25



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12. An object moves along a straight line so that at any time t its acceleration is given by $a(t) = 6t$. At time $t = 0$, the object's velocity is 10 and the object's position is 7. What is the object's position at time $t = 2$?
- (A) 22
(B) 27
(C) 28
(D) 35

Answer D

This option is correct. Velocity is the antiderivative of acceleration and position is the antiderivative of velocity. In each case, the object's velocity and position at $t = 0$ can be used to find the appropriate

" $+ C$ " after each antidifferentiation. The last step is to then evaluate the position function at $t = 2$.

$$a(t) = 6t$$

$$v(t) = 3t^2 + C_1$$

$$v(0) = 3(0)^2 + C_1 = 10$$

$$C_1 = 10$$

$$v(t) = 3t^2 + 10$$

$$p(t) = t^3 + 10t + C_2$$

$$p(0) = (0)^3 + 10(0) + C_2 = 7$$

$$C_2 = 7$$

$$p(t) = t^3 + 10t + 7$$

$$p(2) = 2^3 + 10(2) + 7 = 35$$

13.  A particle travels along a straight line with velocity $v(t) = 3e^{-t/2} \sin(2t)$ meters per second. What is the total distance, in meters, traveled by the particle during the time interval $0 \leq t \leq 2$ seconds?
- (A) 0.835
(B) 1.850
(C) 2.055
(D) 2.261

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14. An object moves along a straight line so that at any time t its acceleration is given by $a(t) = 6t$. At time $t = 0$, the object's velocity is 10 and the object's position is 7. What is the object's position at time $t = 2$?
- (A) 22
(B) 27
(C) 28
(D) 35

**Answer D**

This option is correct. Velocity is the antiderivative of acceleration and position is the antiderivative of velocity. In each case, the object's velocity and position at $t = 0$ can be used to find the appropriate "+ C" after each antidifferentiation. The last step is to then evaluate the position function at $t = 2$.

$$a(t) = 6t$$

$$v(t) = 3t^2 + C_1$$

$$v(0) = 3(0)^2 + C_1 = 10 \Rightarrow C_1 = 10$$

$$v(t) = 3t^2 + 10$$

$$p(t) = t^3 + 10t + C_2$$

$$p(0) = (0)^3 + 10(0) + C_2 = 7 \Rightarrow C_2 = 7$$

$$p(t) = t^3 + 10t + 7$$

$$p(2) = 2^3 + 10(2) + 7 = 35$$

15. An object moves along a straight line so that at any time $t \geq 0$ its velocity is given by $v(t) = 2\cos(3t)$. What is the distance traveled by the object from $t = 0$ to the first time that it stops?
- (A) 0
(B) $\frac{\pi}{6}$
(C) $\frac{2}{3}$
(D) $\frac{\pi}{3}$
(E) $\frac{4}{3}$

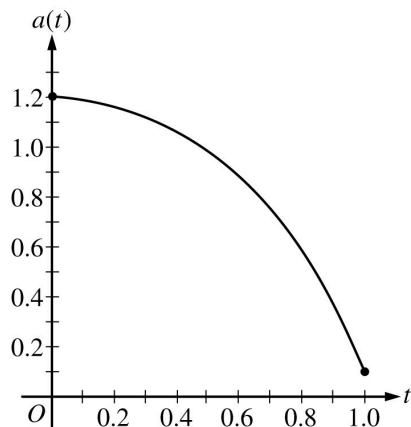


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16. The velocity of a particle moving along the x -axis is given by $v(t) = \sin(2t)$ at time t . If the particle is at $x = 4$ when $t = 0$, what is the position of the particle when $t = \frac{\pi}{2}$?
- (A) 2
(B) 3
(C) 4
(D) 5 ✓
(E) 6
17. A particle travels along the x -axis so that at time $t \geq 0$ its velocity is given by $v(t) = t^2 - 6t + 8$. What is the total distance the particle travels from $t = 0$ to $t = 3$?
- (A) 1
(B) 6
(C) $20/3$
(D) $22/3$ ✓
(E) 8
18.  The velocity of a particle moving along the x -axis is given by $v(t) = 2 - t^2 \sin t$ for $0 \leq t \leq 2$. What is the total distance traveled by the particle between $t = 0$ and $t = 2$?
- (A) -3.637
(B) -1.973
(C) 1.531
(D) 2.539 ✓
(E) 3.637
19.  A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 5te^{-t} - 1$. At $t = 0$, the particle is at position $x = 1$. What is the total distance traveled by the particle from $t = 0$ to $t = 4$?
- (A) 0.366
(B) 0.542
(C) 1.542
(D) 1.821 ✓
(E) 2.821

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20.

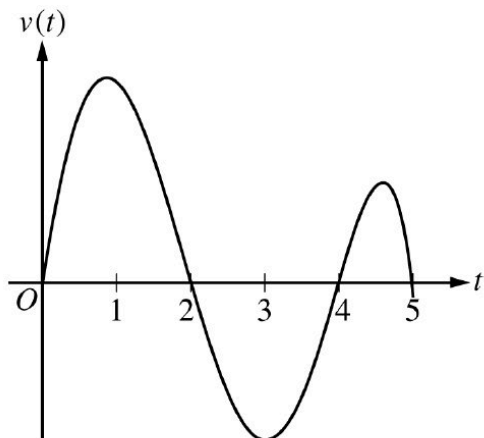


A particle moves along the x -axis so that its acceleration $a(t)$ is given by the graph above for all values of t where $0 \leq t \leq 1$. At time $t = 0$, the velocity of the particle is $-\frac{1}{2}$. Which of the following statements must be true?

- (A) The particle passes through $x = 0$ for some t between $t = 0$ and $t = 1$.
- (B) The velocity of the particle is 0 for some t between $t = 0$ and $t = 1$. ✓
- (C) The velocity of the particle is negative for all values of t between $t = 0$ and $t = 1$.
- (D) The velocity of the particle is positive for all values of t between $t = 0$ and $t = 1$.
- (E) The velocity of the particle is less than $-\frac{1}{2}$ for all values of t between $t = 0$ and $t = 1$.

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21.



Over the time interval $0 \leq t \leq 5$, a particle moves along the x -axis. The graph of the particle's velocity, v , is shown above. Over the time interval $0 \leq t \leq 5$, the particle's displacement is 3 and the particle travels a total distance of 13. What is the value of $\int_2^4 v(t) dt$?

(A) -10

(B) -5

(C) 5

(D) 10

**Answer B**

This option is correct. Let the total area above the t -axis be A and the area below the t -axis be B . The particle's total distance on the time interval $0 \leq t \leq 5$ is:

$$\int_0^5 |v(t)| dt = A + B = 13,$$

and the particle's displacement is:

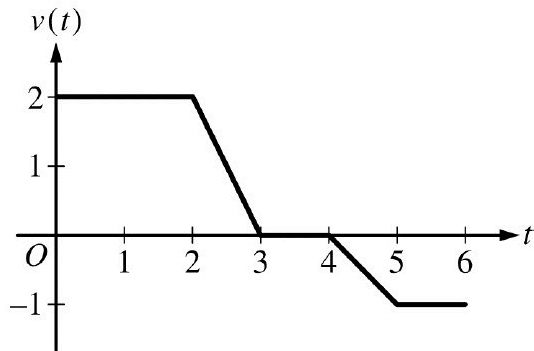
$$\int_0^5 v(t) dt = A - B = 3.$$

Solving the system gives $A = 8$ and $B = 5$. Since the graph of $v(t)$ is below the axis for $2 < t < 4$,

$$\int_2^4 v(t) dt = -B = -5.$$

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22.



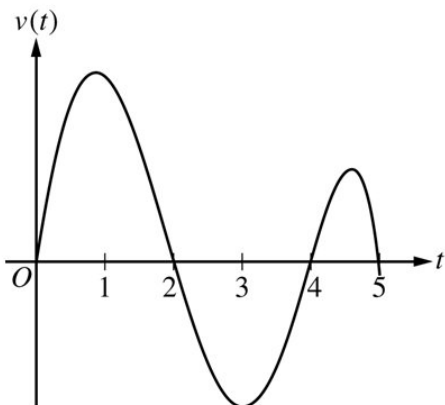
A particle moves along a straight line for 6 seconds so that its velocity, in centimeters per second, is modeled by the graph shown. During the time interval $0 \leq t \leq 6$, what is the total distance the particle travels?

- (A) 2 cm
- (B) 3.5 cm
- (C) 4 cm
- (D) 6.5 cm
- (E) 8.5 cm



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23.



Over the time interval $0 \leq t \leq 5$, a particle moves along the x -axis. The graph of the particle's velocity, v , is shown above. Over the time interval $0 \leq t \leq 5$, the particle's displacement is 3 and the particle travels a total distance of 13. What is the value of $\int_2^4 v(t) dt$?

(A) -10

(B) -5

(C) 5

(D) 10

**Answer B**

This option is correct. Let the total area above the t -axis be A and the area below the t -axis be B . The particle's total distance on the time interval $0 \leq t \leq 5$ is:

$$\int_0^5 |v(t)| dt = A + B = 13,$$

and the particle's displacement is:

$$\int_0^5 v(t) dt = A - B = 3.$$

Solving the system gives $A = 8$ and $B = 5$. Since the graph of $v(t)$ is below the axis for $2 < t < 4$,

$$\int_2^4 v(t) dt = -B = -5.$$

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24. A particle moves along a straight line so that at time $t > 0$ the position of the particle is given by $s(t)$, the velocity is given by $v(t)$, and the acceleration is given by $a(t)$. Which of the following expressions gives the average velocity of the particle on the interval $[2, 8]$?

(A) $\frac{1}{6} \int_2^8 a(t) dt$

(B) $\frac{1}{6} \int_2^8 s(t) dt$

(C) $\frac{s(8) - s(2)}{6}$



(D) $\frac{v(8) - v(2)}{6}$

(E) $v(8) - v(2)$

25. For time $t \geq 0$, the position of a particle traveling along a line is given by a differentiable function s . If s is increasing for $0 \leq t < 2$ and s is decreasing for $t > 2$, which of the following is the total distance the particle travels for $0 \leq t \leq 5$?

(A) $s(0) + \int_0^2 s'(t) dt - \int_2^5 s'(t) dt$

(B) $s(0) + \int_2^5 s'(t) dt - \int_0^2 s'(t) dt$

(C) $\int_2^5 s'(t) dt - \int_0^2 s'(t) dt$

(D) $\left| \int_0^5 s'(t) dt \right|$

(E) $\int_0^5 |s'(t)| dt$



8-2

26. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Mary and Chance walk in the same direction along a straight path. For $0 \leq t \leq 20$, Mary's velocity at time t is given by $M(t) = \frac{6010}{t^2 - 3t + 50.5}$ and Chance's velocity at time t is given by $C(t) = 8.5t^3 e^{-0.45t}$. Both $M(t)$ and $C(t)$ are positive for $0 \leq t \leq 20$ and are measured in meters per minute, and t is measured in minutes. Mary is 12 meters ahead of Chance at time $t = 0$, and Mary remains ahead of Chance for $0 < t \leq 20$.

- (a) Find the value of $\frac{1}{10} \int_5^{15} M(t) \, dt$. Using correct units, interpret the meaning of $\frac{1}{10} \int_5^{15} M(t) \, dt$ in the context of the problem.
- (b) At time $t = 10$, is Mary speeding up or slowing down? Give a reason for your answer.
- (c) Is the distance between Mary and Chance at time $t = 18$ increasing or decreasing? Give a reason for your answer.
- (d) What is the maximum distance between Mary and Chance over the time interval $0 \leq t \leq 20$? Justify your answer.

Part A

For the second point, the response must include reference to average value over the correct time interval with correct units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ value
- ☐ meaning

Solution:

$$\frac{1}{10} \int_5^{15} M(t) dt = 54.409167$$

Mary's average velocity over the time interval $5 \leq t \leq 10$ is 54.409 meters per minute.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses $M(10)$ and $M'(10)$
- ☐ answer with reason

Solution:

$$M(10) > 0 \text{ and } M'(10) = -7.036380 < 0$$

At time $t = 10$, since Mary has positive velocity and negative acceleration, she is slowing down.

Part C

For the point, $M(18) - C(18) > 0$ is required with or without numerical values.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

8-2

Mary's velocity at time $t = 18$ is $M(18) = 18.751950$. Chance's velocity at time $t = 18$ is $C(18) = 15.047042$.

Since $M(18) - C(18) > 0$, the distance between Mary and Chance at time $t = 18$ is increasing.

Part D

The third point is earned with correct use of the Candidates Test.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ considers $M(t) - C(t) = 0$
- ☐ $t = 4.584$ and $t = 16.715$ as candidates
- ☐ justification
- ☐ answer

Solution:

The distance between Mary and Chance at time t is given by $12 + \int_0^t (M(x) - C(x)) dx$.

$$\frac{d}{dt} \left(12 + \int_0^t (M(x) - C(x)) dx \right) = M(t) - C(t) = 0$$

$$\Rightarrow t_1 = 4.583625 \text{ and } t_2 = 16.714620$$

t	Distance between Mary and Chance
0	12
t_1	365.527469
t_2	13.344315
20	27.010082

The maximum distance between Mary and Chance is 365.527 meters.

8-2

27. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Gwen runs back and forth along a straight track. During the time interval $0 \leq t \leq 45$ seconds, Gwen's velocity, in feet per second, is modeled by the function v given by $v(t) = \frac{250 \sin\left(\frac{t^2}{120}\right)}{t+6}$.

- What is the first time, t_1 , that Gwen changes direction? Find Gwen's average velocity over the time interval $0 \leq t \leq t_1$.
- At time $t = 45$, how far is Gwen from her position at time $t = 0$?
- Find the total distance that Gwen runs during the 45 seconds from time $t = 0$ to time $t = 45$.
- Let $a(t)$ represent Gwen's acceleration during the time interval $0 < t < 45$ seconds. Find the value of $\int_{15}^{30} a(t) \, dt$. Using correct units, explain the meaning of $\int_{15}^{30} a(t) \, dt$ in the context of the problem.

Part A

For the first point, supporting work stating that $v(t)$ changes sign is required.

For the second point, supporting work that includes the average value expression with an appropriate definite integral is required. To be eligible for the second point, the presented value of t_1 may be incorrect but must be positive.

The third point requires the first two points to have been earned.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-2



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $t_1 = 19.416$
- ☐ average value integral
- ☐ answer

Solution:

Gwen changes direction when $v(t)$ changes sign. The first time this happens is at $t_1 = 19.416$.

$$\frac{1}{t_1 - 0} \int_0^{t_1} v(t) dt = 7.264377$$

Gwen's average velocity over the time interval $0 \leq t \leq t_1$ is 7.264 feet per second.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{45} v(t) dt = 116.537972$$

At time $t = 45$, Gwen is 116.538 (or 116.537) feet from her position at time $t = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-2



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{45} |v(t)| dt = 250.099483$$

During the 45 seconds from time $t = 0$ to time $t = 45$, the total distance that Gwen runs is 250.099 feet.

Part D

The second point requires 3 things: (1) net change in Gwen's velocity, (2) units of feet per second, and (3) reference to time interval from $t = 15$ to $t = 30$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ interpretation with units

Solution:

$$\int_{15}^{30} a(t) dt = \int_{15}^{30} v'(t) dt = v(30) - v(15) = -4.844 \text{ feet per second}$$

Since acceleration is the rate of change of velocity, $\int_{15}^{30} a(t) dt$ is the net change in Gwen's velocity, in feet per second, during the 15 seconds from time $t = 15$ to time $t = 30$.

8-2

28. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Emma and Lily jog in the same direction along a straight track. For $0 \leq t \leq 15$, Emma's velocity at time t is given by $E(t) = \frac{7510}{t^2 - 7t + 80.22}$ and Lily's velocity at time t is given by $L(t) = 12t^3 e^{-0.5t}$. Both $E(t)$ and $L(t)$ are positive for $0 \leq t \leq 15$ and are measured in meters per minute, and t is measured in minutes. Emma is 10 meters ahead of Lily at time $t = 0$, and Emma remains ahead of Lily for $0 < t \leq 15$.

- (a) Find the value of $\frac{1}{6} \int_2^8 E(t) dt$. Using correct units, interpret the meaning of $\frac{1}{6} \int_2^8 E(t) dt$ in the context of the problem.
- (b) At time $t = 5$, is Emma speeding up or slowing down? Give a reason for your answer.
- (c) Is the distance between Emma and Lily at time $t = 7$ increasing or decreasing? Give a reason for your answer.
- (d) What is the maximum distance between Emma and Lily over the time interval $0 \leq t \leq 15$? Justify your answer.

Part A

For the second point, the response must include reference to average value over the correct time interval with correct units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

8-2

The student response accurately includes both of the criteria below.

- ☐ value
- ☐ meaning

Solution:

$$\frac{1}{6} \int_2^8 E(t) dt = 103.178012$$

Emma's average velocity over the time interval $2 \leq t \leq 8$ is 103.178 meters per minute.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses $E(5)$ and $E'(5)$
- ☐ answer with reason

Solution:

$$E(5) > 0 \text{ and } E'(5) = -4.569193 < 0$$

At time $t = 5$, since Emma has positive velocity and negative acceleration, she is slowing down.

Part C

For the first point, $E(7) - L(7) < 0$ is required with or without numerical values.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

8-2

Emma's velocity at $t = 7$ is $E(7) = 93.617552$.

Lily's velocity at time $t = 7$ is $L(7) = 124.292430$.

Since $E(7) - L(7) < 0$, the distance between Emma and Lily at $t = 7$ is decreasing.

Part D

The third point is earned with correct use of the Candidates Test.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ considers $E(t) - L(t) = 0$
- ☐ $t = 4.233$ and $t = 11.671$ as candidates
- ☐ justification
- ☐ answer

Solution:

The distance between Emma and Lily at time t is given by $10 + \int_0^t (E(x) - L(x)) dx$.

$$\frac{d}{dt} \left(10 + \int_0^t (E(x) - L(x)) dx \right) = E(t) - L(t) = 0$$

$$\Rightarrow t_1 = 4.232614 \text{ and } t_2 = 11.671403$$

t	Distance between Emma and Lily
0	10
t_1	266.998916
t_2	126.880323
15	156.143676

The maximum distance between Emma and Lily is 267 meters.

8-2

29. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Diego runs back and forth along a straight track. During the time interval $0 \leq t \leq 30$ seconds, Diego's velocity, in feet per second, is modeled by the function v given by $v(t) = \frac{200 \sin\left(\frac{t^2}{75}\right)}{t+1}$.

- What is the first time, t_1 , that Diego changes direction? Find Diego's average velocity over the time interval $0 \leq t \leq t_1$.
- At time $t = 30$, how far is Diego from his position at time $t = 0$?
- Find the total distance that Diego runs during the 30 seconds from time $t = 0$ to time $t = 30$.
- Let $a(t)$ represent Diego's acceleration during the time interval $0 < t < 30$ seconds. Using correct units, explain the meaning of $\int_5^{10} a(t) \, dt$ in the context of the problem. Find the value of $\int_5^{10} a(t) \, dt$.

Part A

For the first point, supporting work stating that $v(t)$ changes sign is required.

For the second point, supporting work that includes the average value expression with an appropriate definite integral is required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

8-2

- ☐ $t_1 = 15.350$
- ☐ average value integral
- ☐ answer

Solution:

Diego changes direction when $v(t)$ changes sign. The first time this happens is at $t_1 = 15.350$ (or 15.349).

$$\frac{1}{t_1 - 0} \int_0^{t_1} v(t) dt = 10.503965$$

Diego's average velocity over the time interval $0 \leq t \leq t_1$ is 10.504 (or 10.503) feet per second.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{30} v(t) dt = 128.322555$$

At time $t = 30$, Diego is 128.323 (or 128.322) feet from his position at time $t = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

8-2

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{30} |v(t)| dt = 243.425185$$

During the 30 seconds from time $t = 0$ to time $t = 30$, the total distance that Diego runs is 243.425 feet.

Part D

The first point requires 3 things: (1) net change in Diego's velocity, (2) units of feet per second, and (3) reference to time interval from $t = 5$ to $t = 10$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ interpretation with units
- ☐ answer

Solution:

Since acceleration is the rate of change of velocity, $\int_5^{10} a(t) dt$ is the net change in Diego's velocity, in feet per second, during the 5 seconds from time $t = 5$ to time $t = 10$.

$$\int_5^{10} a(t) dt = \int_5^{10} v'(t) dt = v(10) - v(5) = 6.765 \text{ feet per second}$$

8-2

30. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	2	3	4
$B(t)$ (miles per hour)	1	8	1.5	-5	11

Brandon and Chloe ride their bikes for 4 hours along a flat, straight road. Brandon's velocity, in miles per hour, at time t hours is given by a differentiable function B for $0 \leq t \leq 4$. Values of $B(t)$ for selected times t are given in the table above. Chloe's velocity, in miles per hour, at time t hours is given by the piecewise function C defined by

$$C(t) = \begin{cases} te^{4-t^2} & \text{for } 0 \leq t \leq 2 \\ 12 - 3t - t^2 & \text{for } 2 < t \leq 4. \end{cases}$$

- How many miles did Chloe travel from time $t = 0$ to time $t = 2$?
- At time $t = 3$, is Chloe's speed increasing or decreasing? Give a reason for your answer.
- Is there a time t , for $0 \leq t \leq 4$, at which Brandon's acceleration is equal to 2.5 miles per hour per hour? Justify your answer.
- Is there a time t , for $0 \leq t \leq 2$, at which Brandon's velocity is equal to Chloe's velocity? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned with a connection to the correct part of the piecewise defined function. $\int_0^2 C(t) dt$ by itself is not sufficient to earn the point.

The second point is earned with $-\frac{1}{2}e^{4-t^2}$ or with $-\frac{1}{2}e^u$ for $u = 4 - t^2$. Limits of integration are part of the third point.

The third point does not require units.

8-2



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ integral
- ☐ antiderivative
- ☐ answer

Solution:

$$\int_0^2 te^{4-t^2} dt = -\frac{1}{2}e^{4-t^2} \Big|_{t=0}^{t=2} = -\frac{1}{2} + \frac{1}{2}e^4$$

Chloe traveled $-\frac{1}{2} + \frac{1}{2}e^4$ miles from time $t = 0$ to time $t = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by showing both $C(3)$ and $C'(3)$ are less than zero or that both have the same sign.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $C(3) < 0$ and $C'(3) < 0$
- ☐ answer with reason

Solution:

$$C(3) = -6 < 0$$

$$C'(3) = -9 < 0$$

Chloe's speed is increasing at time $t = 3$ because her velocity and acceleration have the same sign.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-2

The second point is earned with the justification that B is continuous and a numerical difference quotient equal to 2.5. There is no requirement for the response to indicate that B is differentiable, nor to refer to the Mean Value Theorem by name.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{B(4)-B(0)}{4-0}$
- ☐ answer with justification

Solution:

B is differentiable $\Rightarrow B$ is continuous on $[0, 4]$.

$$\frac{B(4)-B(0)}{4-0} = \frac{11-1}{4-0} = 2.5$$

By the Mean Value Theorem, there is a time t , for $0 < t < 4$, such that $B'(t) = 2.5$ miles per hour per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by comparing B and C , either by a subtraction statement or by comparing $B(t)$ and $C(t)$ for some t in $\{0, 1, 2\}$.

The second point is earned with a justification that includes asserting the continuity of B and C or of $B - C$, and showing that $B(0) > C(0)$ and $B(2) < C(2)$. It is not required for the response to specifically refer to the Intermediate Value Theorem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ considers $B(t) - C(t)$
- ☐ answer with justification

Solution:

B and C are continuous on $[0, 2]$, therefore $B - C$ is continuous on $[0, 2]$.

8-2

$$B(0) - C(0) = 1 - 0 > 0$$

$$B(2) - C(2) = 1.5 - 2 < 0$$

By the Intermediate Value Theorem, there is a time t , for $0 < t < 2$, such that $B(t) - C(t) = 0$, or $B(t) = C(t)$.

8-2

31. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle moves along the x -axis so that its velocity at time t is given by $v(t) = 2t \cdot \cos\left(\frac{t^3}{4}\right)$. The particle is at position $x = 3$ at time $t = 0$.

- (a) For $0 < t < \pi$, find all times t during which the particle is moving to the left. Give a reason for your answer.
- (b) Find the total distance traveled by the particle from time $t = 0$ to time $t = 2$. Show the setup for your calculations.
- (c) Find the acceleration of the particle at time $t = 2$. Show the setup for your calculations. Is the particle speeding up or slowing down at time $t = 2$? Give a reason for your answer.
- (d) Find the rightmost position of the particle for $0 \leq t \leq \pi$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers the sign of $v(t)$ earns the first point.

A response that identifies $t = 1.845$ (or $\sqrt[3]{2\pi}$) and/or $t = 2.661$ (or $\sqrt[3]{6\pi}$) and no other values in the interval $0 < t < \pi$ earns the first point.

A response of only $1.845 < t < 2.661$ earns the first point but not the second point.



0	1	2
---	---	---

8-2

The student response accurately includes **both** of the criteria below.

☐ Considers $v(t) = 0$

☐ Answer with reason

Solution:

The particle is moving to the left when $v(t) < 0$.

$$0 < t < \pi \text{ and } v(t) = 0$$

$$\Rightarrow t = A = 1.845270 \text{ and } t = B = 2.661340$$

The particle is moving left for $1.845 < t < 2.661$ because $v(t) < 0$ in this interval.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for $\int_0^2 |v(t)| dt$ or $\int_0^{1.845} v(t) dt - \int_{1.845}^2 v(t) dt$ or an equivalent expression, with or without the differential.

The second point is earned for an answer of 2.595 regardless of whether the first point was earned.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode,

$$\int_0^2 |v(t)| dt = 3.999391.$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integral

☐ Answer

Solution:

$$\int_0^2 |v(t)| dt = 2.595422$$

8-2

The total distance traveled by the particle from time $t = 0$ to time $t = 2$ is 2.595.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimum work needed to earn the first point is $v'(2) = -11.744$ (or -11.743).

$a(2) = -11.743863$ is not sufficient to earn the first point. The connection $v'(t) = a(t)$ or $v'(2) = a(2)$ must be explicitly made.

A response that writes an incorrect statement such as $v'(t) = a(2)$ or $dv = a = -11.744$ does not earn the first point but is eligible for the second point. Note that a response that presents $v' = a = -11.744$ at $t = 2$ does earn the first point.

A response does not need to find the value of $v(2)$; the sign is sufficient. However, a response that reports an incorrect sign or value of $v(2)$ does not earn the second point.

If a value for $v(2)$ is presented, that value must be accurate to the number of decimals presented, for zero up to three decimal places.

A response must declare a value for $a(2)$ to be eligible to earn the second point. In order to earn the second point, the presented conclusion must be consistent with a negative velocity at time $t = 2$ and the presented value of $a(2)$.

Degree mode: In degree mode, $v'(2) = 1.991472$ and $v(2) = 3.997563$. A response that declares $v'(2) = 1.991472$ will earn the second point in the presence of $v(2) = 3.997563$ or $v(2) > 0$ and the conclusion that the particle is speeding up.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Acceleration with setup
- ☐ Speeding up with reason

Solution:

$$v'(2) = a(2) = -11.743863$$

The acceleration at time $t = 2$ is -11.744 (or -11.743).

$$v(2) = -1.664587 < 0$$

8-2

The particle is speeding up at time $t = 2$ because the particle's velocity and acceleration are both negative at that time.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an attempt to evaluate $x(A)$.

The second point is earned for considering both 0 and π as candidates, verbally or in a table.

To earn the third point the response must identify the rightmost position at 5.471 and either:

- evaluate x at $t = 0$, $t = A$, $t = B$, and $t = \pi$, or
- evaluate x at $t = A$ and $t = \pi$ and eliminate $t = 0$ and $t = B$ from consideration by arguing that $0 < t < A \Rightarrow v(t) > 0$ and $B < t < \pi \Rightarrow v(t) > 0$.

All evaluations must be correct for the number of digits presented for one up to three decimal places.

Degree mode: The response is only eligible to earn the first two points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x(A)$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

$$0 \leq t \leq \pi \text{ and } v(t) = 0$$

$$\Rightarrow t = 0, t = A = 1.845270, \text{ and } t = B = 2.661340$$

Because $v(t)$ is continuous on $[0, \pi]$, the candidates for the time at which the particle is at its rightmost position are $t = 0$, $t = A$, $t = B$, and $t = \pi$.

$$x(0) = 3$$

$$x(A) = 3 + \int_0^A v(t) \, dt = 5.471352$$

8-2

$$x(B) = 3 + \int_0^B v(t) \, dt = 3.151377$$

$$x(\pi) = 3 + \int_0^\pi v(t) \, dt = 4.972567$$

The rightmost position of the particle for $0 \leq t \leq \pi$ is $x = 5.471$.

8-2

32. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The position of a particle moving along the x -axis at time t is given by the twice-differentiable function $x(t)$, where $x(0) = 2$. For $0 \leq t \leq 5$, the derivative of $x(t)$ is given by $x'(t) = \sin(t^{1.1})\sqrt{10t+4}$.

- (a) Find all times t in the open interval $0 < t < 5$ when the particle changes direction. Give a reason for your answer.
- (b) Find the total distance traveled by the particle from time $t = 1$ to time $t = 4$.
- (c) Find the acceleration of the particle at time $t = 4$. Is the speed of the particle increasing or decreasing at time $t = 4$? Give a reason for your answer.
- (d) Find the position of the particle at time $t = 5$. For $0 \leq t \leq 5$, is there a time t when the particle is at the origin? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point cannot be earned implicitly by reporting $t = 2.831095$; the setup must be shown.

A response that finds an incorrect value of t does not earn the second point.

A response may find and report extraneous solutions to the equation $x'(t) = 0$. If the response does not specifically eliminate the extraneous solutions, it will not earn the second point.

The second point can be earned by applying the Second Derivative Test. This includes reporting that $x'(t) = 0$ at $t = 2.831$, and that $x''(2.831) < 0$, and therefore the particle changes direction when $t = 2.831$. Such a response need not report the value of $x''(2.831)$, but if a value is presented it must be correct for the presented digits, up to three decimal places ($x''(2.831) = -6.938$).

8-2

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $t = 112.26587$, which is not in $0 < t < 5$. Therefore, such a response does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Sets $x'(t) = 0$
- ☐ Answer with reason

Solution:

$$x'(t) = 0 \Rightarrow t = 2.831095$$

The particle changes direction when $t = 2.831$ because the velocity, $x'(t)$, changes from positive to negative at this value.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents incorrect limits or no limits does not earn the first point, but, in the presence of the correct answer, does earn the second point.

A response that omits the absolute value on the integrand or uses $x(t)$ for the integrand does not earn the first point, but, in the presence of the correct answer, does earn the second point.

A response that presents $\left| \int_1^4 x'(t) dt \right|$ does not earn the first point, but, in the presence of the correct answer, does earn the second point.

Degree mode: In degree mode, $\int_1^4 |x'(t)| dt = 0.813$ (or 0.812). (See degree mode statement in part (a).)



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integral(s)

8-2

☐ Answer
Solution:

$$\int_1^4 |x'(t)| dt = 10.5556$$

The total distance traveled by the particle from time $t = 1$ to time $t = 4$ is 10.556 (or 10.555).

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimum work needed to earn the first point is $x''(4) = -1.732$ (or -1.731).

$a(4) = -1.732$ (or -1.731) is not sufficient to earn the first point. The connection $a(4) = x''(4)$, or $a(t) = x''(t)$, must be explicitly made.

A response does not need to find the value of $v(4)$; the sign is sufficient. However, a response that reports an incorrect sign or value of $v(4)$ does not earn the second point. A response must declare a value for $x''(4)$ to be eligible for the second point.

If a value for $v(4)$ is presented, that value must be accurate for the digits shown, up to three decimal places.

A response that declares an incorrect value of $x''(4)$ is eligible for the second point. The conclusion must be consistent with the presented value of $x''(4)$ and the correct sign/value of $v(4)$.

Degree mode: In degree mode, $x''(4) = a(4) = 0.206$ (or 8.415) and $x'(4) = v(4) = 0.531 > 0$. (See degree mode statement in part (a).)



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $x''(4)$
☐ Conclusion with reason

Solution:

$$x''(4) = a(4) = -1.732 \text{ (or } -1.731)$$

$$x'(4) = v(4) = -6.587438$$

8-2

Because the values of the velocity and acceleration have the same sign at $t = 4$, the speed is increasing at time $t = 4$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents an integral of $\int_0^5 |x'(t)| dt$ does not earn the first point.

- This response will earn the second point only for a value of $x(5) = 2 + \int_0^5 |x'(t)| dt = 19.612$.

- If this response earns the second point, it will earn the third point only for reasoning that there is not sufficient information to determine whether there is a time t , for $0 \leq t \leq 5$, when the particle is at the origin. The reasoning must reference the Intermediate Value Theorem.

To earn the third point, the response must include:

- differentiability implies continuity (the continuity of $x(t)$ by itself is insufficient),
- $x(5) < 0 < x(0)$ or -0.566 (or -0.565) $< 0 < 2$ (this may be expressed in words or symbols), and
- “yes.”

A response of $x(5) = \int_0^5 x'(t) dt = -2.566$ (or -2.565) earns the first point, does not earn the second point, and is eligible to earn the third point.

A response with incorrect or no limits of integration does not earn the first point. Such a response that presents the correct value of $x(5)$ does earn the second point, and is eligible to earn the third point. A response with incorrect or no limits that presents an incorrect value of $x(5)$ will not earn any points.

A response that solves the equation $2 + \int_0^t x'(s) ds = 0$ for t and reports a value of $t = 4.841$ earns the third point, but must correctly find $x(5)$ in order to earn the first 2 points.

Degree mode: In degree mode, $x(5) = 2 + \int_0^5 x'(t) dt = 3.479$. (See degree mode statement in part (a).)



0	1	2	3
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The student response accurately includes **one** of the criteria below.

☐ $\int_0^5 x'(t) dt$

8-2

- ☐ $x(5)$
- ☐ Answer with justification

$$x(5) = 2 + \int_0^5 x'(t) dt = -0.565953$$

The position of the particle at time $t = 5$ is -0.566 (or -0.565).

The particle is at the origin when $x(t) = 0$.

$x(t)$ is twice-differentiable. $\Rightarrow x(t)$ is continuous.

Because $x(5) < 0 < x(0) = 2$ and $x(t)$ is continuous, by the Intermediate Value Theorem there must be a time t in $0 \leq t \leq 5$ when the particle is at the origin.

33. A particle moves along the x -axis with velocity given by $v(t)$ for time $t \geq 0$. The particle is at position $x = 7$ at time $t = 2$. Which of the following gives the position of the particle at time $t = 5$?

- (A) $\int_0^5 v(t) dt$
- (B) $\int_2^5 v(t) dt$
- (C) $7 + \int_0^5 v(t) dt$
- (D) $7 + \int_2^5 v(t) dt$ ✓

Answer D

Correct. For a particle in rectilinear motion over an interval of time, the definite integral of velocity represents the particle's displacement over the interval of time. Since the initial position of the particle at time $t = 2$ is $x = 7$, the position at time $t = 5$ would be $x(5) = x(2) + \int_2^5 v(t) dt = 7 + \int_2^5 v(t) dt$.

8-2

34.  The velocity, in centimeters per second, of a particle moving along a line is given by $v(t) = t^3 - \frac{11}{7}t$. What is the total distance, in centimeters, traveled by the particle from $t = 0$ seconds to $t = 2$ seconds?
- (A) 0.857
- (B) 2.092
- (C) 3.165
- (D) 4.857

Answer B

Correct. The definite integral of speed represents the particle's total distance traveled over an interval of time.

$$\int_0^2 |v(t)| dt = \int_0^2 \left| t^3 - \frac{11}{7}t \right| dt = 2.092$$

The graphing calculator is used to evaluate the definite integral.

35.  A particle travels along the x -axis so that its velocity at time $t \geq 0$ is given by $\frac{dx}{dt} = \sin(t^2)$. What is the total distance the particle travels from $t = 0$ to $t = 2.7$?
- (A) 0.519
- (B) 0.845
- (C) 1.448
- (D) 2.887

Answer C

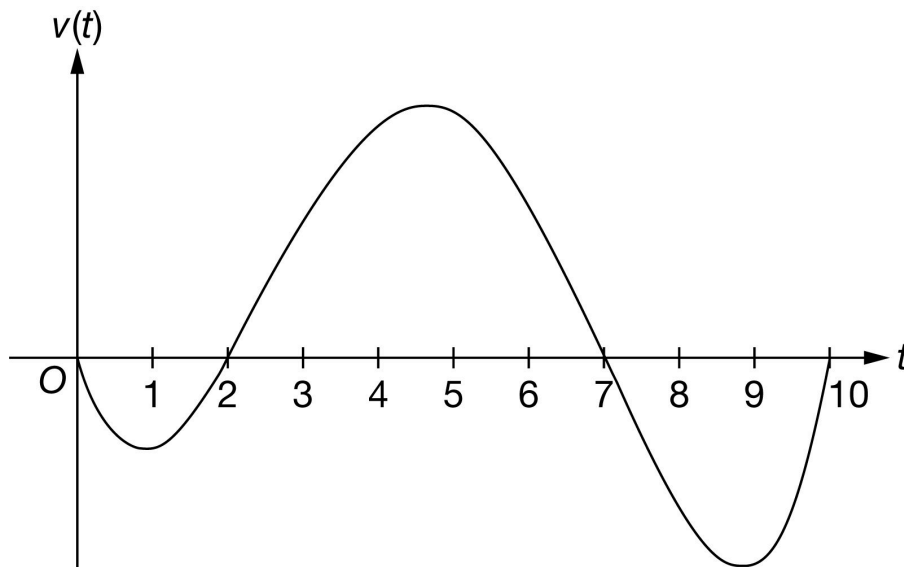
Correct. The definite integral of speed represents the particle's total distance traveled over an interval of time.

$$\int_0^{2.7} |v(t)| dt = \int_0^{2.7} |\sin(t^2)| dt = 1.448$$

The definite integral is evaluated on the calculator.

8-2

36.



Over the time interval $0 \leq t \leq 10$, a particle moves along the x -axis. The graph of the particle's velocity $v(t)$ at time t is shown above. Over the time interval $0 \leq t \leq 10$, the particle's displacement is 15 and the particle travels a total distance of 73. What is the total distance that the particle travels while moving to the left?

(A) 15

(B) 29

(C) 44

(D) 58



Answer B

Correct. Let A be the area bounded by the graph of $v(t)$ and the t -axis that is above the t -axis, and let B be the total bounded area that is below the t -axis. The particle's displacement over the time interval $0 \leq t \leq 10$ is

$\int_0^{10} v(t) dt = A - B = 15$. The particle's total distance is $\int_0^{10} |v(t)| dt = A + B = 73$. Solving this system

gives $A = 44$ and $B = 29$. The particle is moving to the left when $v(t) < 0$, which happens over the time intervals $0 < t < 2$ and $7 < t < 10$. The total distance that the particle travels over these two time intervals is therefore

$$\int_0^2 |v(t)| dt + \int_7^{10} |v(t)| dt = B = 29.$$

8-2

37. The velocity of a particle moving along the x -axis is given by $v(t) = e^{t-1} - 1$. Which of the following gives the total distance traveled by the particle over the time interval $0 \leq t \leq 3$?

(A) $|v(0)| \cdot 3$

(B) $\int_0^1 v(t) dt - \int_1^3 v(t) dt$

(C) $\int_0^3 v(t) dt$

(D) $-\int_0^1 v(t) dt + \int_1^3 v(t) dt$

**Answer D**

Correct. The particle's total distance over the time interval $[0, 3]$ is $\int_0^3 |v(t)| dt$. To evaluate this integral, first determine where the velocity changes sign on the interval $[0, 3]$. At $t = 1$, $v(t) = e^{(1-1)} - 1 = e^0 - 1 = 0$. $v(t) \leq 0$ for $t \leq 1$, and $v(t) \geq 0$ for $t \geq 1$.

$$\int_0^3 |v(t)| dt = \int_0^1 (1 - e^{t-1}) dt + \int_1^3 (e^{t-1} - 1) dt$$

38.  For time $t \geq 0$, the velocity of a particle moving along the x -axis is given by $v(t) = \cos(8e^{-0.2t})$. The initial position of the particle at time $t = 0$ is $x = 2.5$. What is the displacement of the particle from time $t = 0$ to time $t = 15$?

(A) 1.067

(B) 2.526

(C) 5.026

(D) 9.243

**Answer B**

Correct. The particle's displacement over the time interval $0 \leq t \leq 16$ is found by integrating the velocity over the interval $0 \leq t \leq 16$. The calculator is used to do the numerical integration.

$$\int_0^{15} v(t) dt = \int_0^{15} \cos(8e^{-0.2t}) dt = 2.526$$

8-2

39. For time $t \geq 0$, the acceleration of a particle moving along the x -axis is given by $a(t) = 12e^{-2t} + 18t$. At time $t = 0$, the velocity of the particle is 2 and the position of the particle is 4. What is the position of the particle at time $t = 2$?
- (A) $-6e^{-4} + 44$
- (B) $3e^{-4} + 24$
- (C) $3e^{-4} + 41$ ✓
- (D) $12e^{-4} - 4$

Answer C

Correct. Velocity is the antiderivative of acceleration, and position is the antiderivative of velocity. In each case, the particle's velocity and position at $t = 0$ can be used to find the appropriate “ $+C$ ” after each antidifferentiation. The last step is to then evaluate the position function at $t = 2$.

$$\frac{dv}{dt} = a(t) = 12e^{-2t} + 18t, \quad v(0) = 2$$

$$v(t) = -6e^{-2t} + 9t^2 + C_1$$

$$v(0) = -6 + C_1 = 2 \Rightarrow C_1 = 8$$

$$\frac{dx}{dt} = v(t) = -6e^{-2t} + 9t^2 + 8, \quad x(0) = 4$$

$$x(t) = 3e^{-2t} + 3t^3 + 8t + C_2$$

$$x(0) = 3 + C_2 = 4 \Rightarrow C_2 = 1$$

$$x(t) = 3e^{-2t} + 3t^3 + 8t + 1$$

$$x(2) = 3e^{-4} + 24 + 16 + 1 = 3e^{-4} + 41$$

8-2

40. For time $t \geq 0$, the velocity of a particle moving along the x -axis is given by $v(t) = \sin t$. The initial position of the particle at time $t = 0$ is $x = 5$. Which of the following gives the total distance the particle travels from time $t = 0$ to time $t = 3\pi$?

(A) $\int_0^{3\pi} \sin t \, dt$

(B) $5 + \int_0^{3\pi} |\sin t| \, dt$

(C) $\int_0^{3\pi} |\sin t| \, dt$ ✓

(D) $5 + \int_0^{3\pi} \sin t \, dt$

Answer C

Correct. The particle's total distance over the time interval $0 \leq t \leq 3\pi$ is found by integrating the speed, $|v(t)| = |\sin t|$, over the interval $0 \leq t \leq 3\pi$.

41. A particle moves along the x -axis with velocity given by $v(t) = (t - 1)e^{1-t}$ for time $t \geq 0$. If the particle is at position $x = 3$ at time $t = 0$, which of the following gives the position of the particle at time $t = 2$?

(A) $\int_0^2 (t - 1)e^{1-t} \, dt$

(B) $3 + \int_0^2 (t - 1)e^{1-t} \, dt$ ✓

(C) $\left. \frac{d}{dt} ((t - 1)e^{1-t}) \right|_{t=2}$

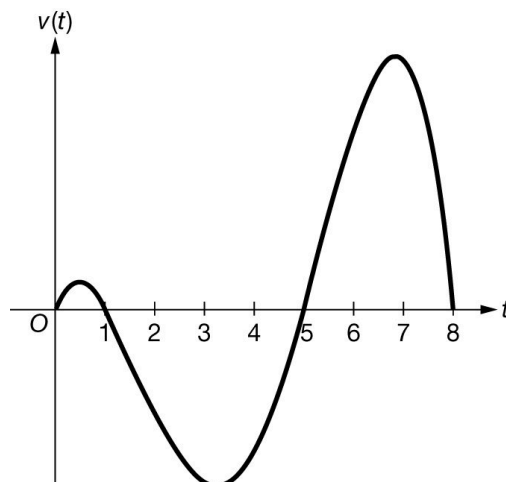
(D) $\int_0^2 |(t - 1)e^{1-t}| \, dt$

Answer B

Correct. The definite integral of the velocity from $t = 0$ to $t = 2$ is the particle's net change in position over the time interval $0 \leq t \leq 2$. Adding the particle's initial position to its net change in position gives the position of the particle at time $t = 2$.

8-2

42.



Over the time interval $0 \leq t \leq 8$, a particle moves along the x -axis. The graph of the particle's velocity at time t , $v(t)$, is shown in the figure above. Over the time interval $0 \leq t \leq 8$, the particle's displacement is 100 units to the right and the particle travels a total distance of 1875. What is the total distance that the particle travels while moving to the left?

- (A) 100
(B) 887.5
(C) 987.5
(D) 1775

**Answer B**

Correct. Let A be the area bounded by the graph of v and the t -axis that is above the t -axis, and let B be the total bounded area that is below the t -axis. The particle's displacement over the time interval $0 \leq t \leq 8$ is

$\int_0^8 v(t) dt = A - B = 100$. The particle's total distance is $\int_0^8 |v(t)| dt = A + B = 1875$. Solving this system gives $A = 987.5$ and $B = 887.5$. The particle is moving to the left when $v(t) < 0$, which happens over the time interval $1 < t < 5$. The total distance that the particle travels over the time interval is therefore

$$\int_1^5 |v(t)| dt = B = 887.5.$$

8-2

43. The velocity of a particle moving along the x -axis is given by $v(t) = -\sin\left(t - \frac{\pi}{4}\right)$. Which of the following gives the total distance traveled by the particle over the time interval $0 \leq t \leq \pi$?

(A) $|v(0)| \cdot \pi$

(B) $-\int_0^{\frac{\pi}{4}} v(t) dt + \int_{\frac{\pi}{4}}^{\pi} v(t) dt$

(C) $\int_0^{\pi} v(t) dt$

(D) $\int_0^{\frac{\pi}{4}} v(t) dt - \int_{\frac{\pi}{4}}^{\pi} v(t) dt$



Answer D

Correct. The particle's total distance over the time interval $[0, \pi]$ is $\int_0^{\pi} |v(t)| dt$. To evaluate this integral, first determine where the velocity changes sign on the interval. At $t = \frac{\pi}{4}$, $v\left(\frac{\pi}{4}\right) = -\sin\left(\frac{\pi}{4} - \frac{\pi}{4}\right) = 0$. For $0 \leq t \leq \frac{\pi}{4}$, $v(t) \geq 0$, and for $\frac{\pi}{4} \leq t \leq \pi$, $v(t) \leq 0$.

$$\int_0^{\pi} |v(t)| dt = \int_0^{\frac{\pi}{4}} v(t) dt - \int_{\frac{\pi}{4}}^{\pi} v(t) dt$$

44.  For time $t \geq 0$, the velocity of a particle moving along the x -axis is given by $v(t) = \arctan(t^2 + t - 1)$. The initial position of the particle at time $t = 0$ is $x = 0.085$. What is the displacement of the particle from time $t = 0$ to time $t = 5$?

(A) 2.322

(B) 5.504

(C) 5.589

(D) 6.106



Answer B

Correct. The particle's displacement over the time interval $0 \leq t \leq 5$ is found by integrating the velocity over the interval $0 \leq t \leq 5$. The calculator is used to do the numerical integration.

$$\int_0^5 v(t) dt = \int_0^5 \arctan(t^2 + t - 1) dt = 5.504$$

8-2

45. For time $t \geq 0$, the acceleration of a particle moving along the x -axis is given by $a(t) = \cos t + t$. At time $t = 0$, the velocity of the particle is 8 and the position of the particle is 0. What is the position of the particle at time $t = \frac{\pi}{4}$?
- (A) $\frac{\sqrt{2}}{2} + \frac{\pi^2}{32} + 8$
- (B) $-\frac{\sqrt{2}}{2} + \frac{\pi^3}{384}$
- (C) $-\frac{\sqrt{2}}{2} + \frac{\pi^3}{384} + 2\pi + 1$ ✓
- (D) $\frac{\sqrt{2}}{2} + \frac{\pi^3}{384} + 2\pi - 1$

Answer C

Correct. Velocity is the antiderivative of acceleration, and position is the antiderivative of velocity. In each case, the particle's velocity and position at $t = 0$ can be used to find the appropriate “ $+C$ ” after each antidifferentiation. The last step is to then evaluate the position function at $t = \frac{\pi}{4}$.

$$\frac{dv}{dt} = a(t) = \cos t + t, \quad v(0) = 8$$

$$v(t) = \int a(t) dt = \sin t + \frac{t^2}{2} + C_1$$

$$v(0) = 8 \Rightarrow C_1 = 8$$

$$\frac{dx}{dt} = v(t) = \sin t + \frac{t^2}{2} + 8$$

$$x(t) = -\cos t + \frac{t^3}{6} + 8t + C_2$$

$$x(0) = -1 + 0 + 0 + C_2 = 0 \Rightarrow C_2 = 1$$

$$x(t) = -\cos t + \frac{t^3}{6} + 8t + 1$$

$$x\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2} + \frac{\pi^3}{384} + 2\pi + 1$$

8-2

46. For time $t \geq 0$, the velocity of a particle moving along the x -axis is given by $v(t) = x^3 - 3x^2 + 2x$. The initial position of the particle at time $t = 0$ is $x = 4$. Which of the following gives the total distance the particle travels from time $t = 0$ to time $t = 3$?

(A) $\int_0^3 (x^3 - 3x^2 + 2x) \, dx$

(B) $4 + \int_0^3 |x^3 - 3x^2 + 2x| \, dx$

(C) $\int_0^3 |x^3 - 3x^2 + 2x| \, dx$

(D) $4 + \int_0^3 (x^3 - 3x^2 + 2x) \, dx$

Answer C

Correct. The particle's total distance over the time interval $0 \leq t \leq 3$ is found by integrating the speed, $|v(t)| = |x^3 - 3x^2 + 2x|$, over the interval $0 \leq t \leq 3$.

47. A particle moves along the x -axis with velocity given by $v(t) = 4t - 20 \sin(t^2 + 1)$ for time $t \geq 0$. If the particle is at position $x = \pi$ at time $t = 0$, which of the following gives the position of the particle at time $t = 4$?

(A) $\int_0^4 (4t - 20 \sin(t^2 + 1)) \, dt$

(B) $\pi + \int_0^4 (4t - 20 \sin(t^2 + 1)) \, dt$

(C) $\left. \frac{d}{dt} (4t - 20 \sin(t^2 + 1)) \right|_{t=4}$

(D) $\int_0^4 |4t - 20 \sin(t^2 + 1)| \, dt$

Answer B

Correct. The definite integral of the velocity from $t = 0$ to $t = 4$ is the particle's net change in position over the time interval $0 \leq t \leq 4$. Adding the particle's initial position to its net change in position gives the position of the particle at time $t = 4$.

8-2

48. A referee moves along a straight path on the side of an athletic field. The velocity of the referee is given by $v(t) = 4(t - 6) \cos(2t + 5)$, where t is measured in minutes and $v(t)$ is measured in meters per minute.



What is the total distance traveled by the referee, in meters, from time $t = 2$ to time $t = 6$?

- (A) 3.933
(B) 14.578
(C) 21.667
(D) 29.156

**Answer C**

Correct. The referee's total distance traveled on the time interval $2 \leq t \leq 6$ is

$\int_2^6 |v(t)| dt = \int_2^6 |4(t - 6) \cos(2t + 5)| dt = 21.667$, where the numerical integration is done with the calculator.

49.  A particle moves along a straight line with velocity given by $v(t) = \frac{t^2 - 2t}{e^t}$ for time $t \geq 0$. What is the total distance traveled by the particle from time $t = 0$ to the time the particle first changes direction?

- (A) 0.191
(B) 0.461
(C) 0.541
(D) 0.699

**Answer C**

Correct. The particle changes direction when $v(t) = 0$ and v changes sign. $v(t) = 0$ when $0 = t^2 - 2t = t(t - 2)$, so the first (and only time) the particle changes direction is when $t = 2$. The definite integral of speed represents the particle's total distance traveled over an interval of time.

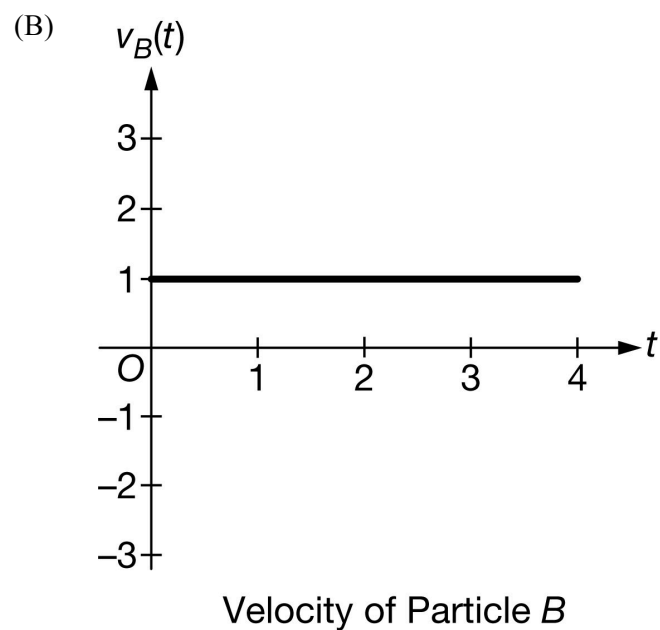
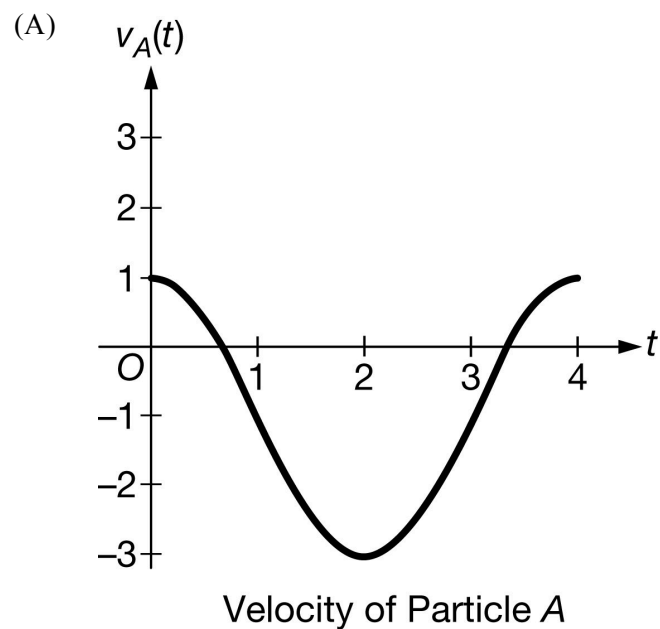
$$\int_0^2 |v(t)| dt = \int_0^2 \left| \frac{t^2 - 2t}{e^t} \right| dt = 0.541$$

The calculator is used to evaluate the definite integral.

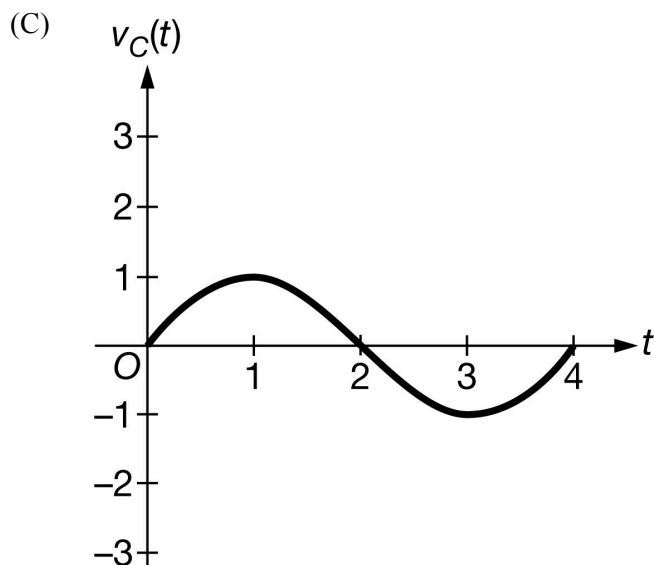
8-2

50. For time $0 \leq t \leq 4$, four particles— A , B , C , and D —move along parallel horizontal lines in the xy -plane. At time $t = 0$, the x -coordinate of the position of each particle is $x = 0$. The graphs of the velocity of each particle, in meters per second, are shown below. Which is the graph of the velocity of the particle that is farthest to the right at time $t = 4$?

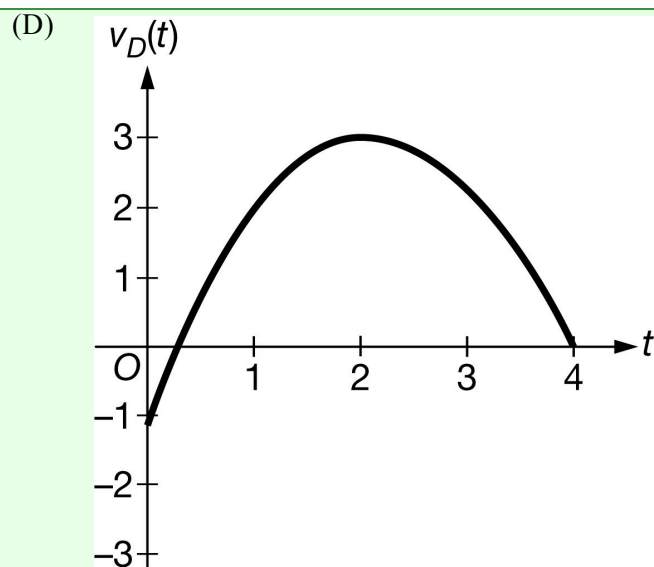
8-2



8-2



Velocity of Particle C



Velocity of Particle D

Answer D

Correct. The x -coordinate of the position of each particle at time $t = 4$ is the value of the definite integral

$\int_0^4 v(t) \, dt$, where $v(t)$ is the particle's velocity. The value of the definite integrals for each particle can be estimated by comparing the areas of the regions bounded by the graph of the particle's velocity function and the t -axis.

$\int_0^4 v_A(t) \, dt < 0$ since the region below the t -axis has more area than the region above, so particle A is to the left of $x = 0$ at time $t = 4$.

8-2

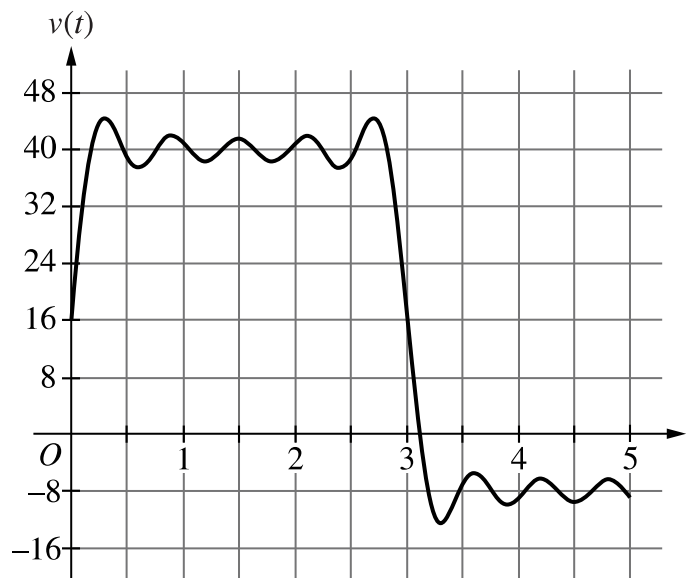
$\int_0^4 v_B(t) dt = 4$ because the region is a rectangle of height 1 and width 4, so particle B is at $x = 4$ at time $t = 4$.

$\int_0^4 v_C(t) dt \approx 0$ since the areas of the regions above and below the t -axis are almost the same, so particle C is approximately at $x = 0$ at time $t = 4$.

$\int_0^4 v_D(t) dt > 4$ because a rectangle of width 2 and height 2 can be placed inside the region in the first quadrant bounded by the graph of v_D and the t -axis, and so particle D is more than 4 meters to the right at time $t = 4$. Therefore particle D is farthest to the right at time $t = 4$.

8-2

51.



For time $t \geq 0$, a particle moves along a straight line with velocity $v(t)$ meters per second. The graph of v is shown. Of the following, which is the best estimate for the total distance, in meters, traveled by the particle from time $t = 0$ to $t = 5$ seconds?

- (A) 24
- (B) 80
- (C) 104
- (D) 136



Answer D

Correct. The particle's total distance on the time interval $0 \leq t \leq 5$ is the definite integral of the particle's speed over that interval, so $\int_0^5 |v(t)| dt$. The value of this definite integral can be approximated by the area bounded by the graph of v and the t -axis, as follows.

$$\int_0^5 |v(t)| dt = \int_0^3 |v(t)| dt + \int_3^5 |v(t)| dt \approx 40(3) + 8(2) = 120 + 16 = 136$$

8-2

52. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Stephen swims back and forth along a straight path in a 50-meter-long pool for 90 seconds. Stephen's velocity is modeled by $v(t) = 2.38e^{-0.02t} \sin\left(\frac{\pi}{56}t\right)$, where t is measured in seconds and $v(t)$ is measured in meters per second.

- (a) Find all times t in the interval $0 < t < 90$ at which Stephen changes direction. Give a reason for your answer.
- (b) Find Stephen's acceleration at time $t = 60$ seconds. Show the setup for your calculations, and indicate units of measure. Is Stephen speeding up or slowing down at time $t = 60$ seconds? Give a reason for your answer.
- (c) Find the distance between Stephen's position at time $t = 20$ seconds and his position at time $t = 80$ seconds. Show the setup for your calculations.
- (d) Find the total distance Stephen swims over the time interval $0 \leq t \leq 90$ seconds. Show the setup for your calculations.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers $v(t) = 0$ earns the first point.

A response of "Stephen changes direction when his velocity changes sign" earns the first point for considering the sign of $v(t)$ but must include the answer of $t = 56$ in order to earn the second point.

A response of $t = 56$ with no supporting work does not earn either point.

Any presented values of outside the interval $0 < t < 90$ will not affect scoring.

8-2



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers sign of $v(t)$
- ☐ Answer with reason

Solution:

For $0 < t < 90$, $v(t) = 0 \Rightarrow t = 56$.

Stephen changes direction when his velocity changes sign. This occurs at $t = 56$ seconds.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimum work needed to earn the first point is $v'(60) = -0.036$.

- $a(60) = -0.0360162$ is not sufficient to earn the first point. The connection $v'(t) = a(t)$ or $v'(60) = a(60)$ must be explicitly shown.

A response must declare a value for $a(60)$ to be eligible for the second point.

In order to earn the third point the presented conclusion must be consistent with a negative velocity at time $t = 60$ and the presented value of $a(60)$.

A response does not need to find the value of $v(60)$; an implied sign is sufficient.

- The statement “Stephen is speeding up because $a(60)$ and $v(60)$ have the same sign” (or equivalent) earns the third point, provided a negative value is presented for $a(60)$.

A response that reports an incorrect sign or value of $v(60)$ does not earn the third point. Any presented value of $v(60)$ must be correct for the number of digits presented, from one up to three decimal places in order to earn the third point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode).

- In degree mode, there are two possible values for $v'(60)$. A response that declares $v'(60) = -0.000141$ does not earn the first point but would earn the third point in the presence of $v(60) = 0.042089$ or $v(60) > 0$ and the conclusion that Stephen is slowing down.

8-2

- Similarly, a response that declares $v'(60) = 0.039304$ does not earn the first point but would earn the third point in the presence of $v(60) = 0.042089$ or $v(60) > 0$ and the conclusion that Stephen is speeding up.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $a(60)$ with setup
- ☐ Acceleration units
- ☐ Speeding up with reason

Solution:

$$v'(60) = a(60) = -0.0360162$$

Stephen's acceleration at time $t = 60$ seconds is -0.036 meter per second per second.

$$v(60) = -0.1595124 < 0$$

Stephen is speeding up at time $t = 60$ seconds because Stephen's velocity and acceleration are both negative at that time.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for $\int_{20}^{80} v(t)dt$ (or the equivalent) with or without the differential.

The second point is earned only for an answer of 23.384 (or 23.383) regardless of whether the first point was earned.

Degree mode: In degree mode, $\int_{20}^{80} v(t)dt = 2.407982$.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

8-2

Solution:

$$\int_{20}^{80} v(t) dt = 23.383997$$

The distance between Stephen's positions at $t = 20$ seconds and $t = 80$ seconds is 23.384 (or 23.383) meters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for $\int_0^{90} |v(t)| dt$ or $\int_0^{56} v(t) dt - \int_{56}^{90} v(t) dt$ (or the equivalent), with or without the differential(s).

The second point is earned only for an answer of 62.164 regardless of whether the first point was earned.

Degree mode: In degree mode, $\int_0^{90} |v(t)| dt = 3.127892$.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

Solution:

$$\int_0^{90} |v(t)| dt = 62.164216$$

The total distance Stephen swims over the time interval $0 \leq t \leq 90$ seconds is 62.164 meters.

8-2

53. A particle is traveling along the x -axis. For time $0 \leq t \leq 5$, the velocity of the particle is defined by $v(t) = 3^{-t^2} - \sqrt{t + 0.4}$. At time $t = 5$, the particle is at position $x = -4$.

**Part A**

Find $a(2)$, the acceleration of the particle at time $t = 2$. Show the setup for your calculations. Is the speed of the particle increasing or decreasing at time $t = 2$? Give a reason for your answer.

Part B

At what time in the interval $0 \leq t \leq 5$ does the particle change direction? Explain your reasoning.

Part C

What is the position of the particle at time $t = 0$? Show the work that leads to your answer.

Part D

Find the total distance traveled by the particle over the time interval $0 \leq t \leq 5$. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for the correct answer with a setup demonstrating the relationship $v' = a$. This could be shown by “ $v'(2) = -0.377$ ” or “ $v' = a$ and $a(2) = -0.377$.” A response of just “ $a(2) = -0.377$ ” is not sufficient to earn **P1**.
- For **P1**, a reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point.
- A response with an unsupported value of -0.377 does not earn **P1**.

Model Solution:

$$a(2) = v'(2) = -0.377001$$

The acceleration of the particle at time $t = 2$ is -0.377 .



0	1
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The response accurately includes the criteria below.

- Acceleration with setup

8-2

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible for **P2**, a response must declare a value of $a(2)$, declare a value of $v'(2)$, or present the value -0.377 . A response with an unsupported value of -0.377 is eligible for **P2**.
- **P2** can only be earned for a response that is consistent with a negative velocity at time $t = 2$ and the presented value of $a(2)$.
- Any presented value of $v(2)$ must be correct to the number of decimal places presented (up to three decimal places) in order to be eligible for **P2**.
- To earn **P2**, the response does not need to present a value of $v(2)$; an implied sign is sufficient. Any statement equivalent to “The speed of the particle is increasing because $a(2)$ and $v(2)$ have the same sign” will earn **P2**, provided the presented value of $a(2)$ is negative.
- Alternate solutions for **P2**:
 - $\text{Speed} = |v(t)|$ and its derivative is positive when $t = 2$, therefore the speed of the particle is increasing.
 - $a(2)v(2) > 0$, therefore the speed of the particle is increasing.

Model Solution:

$$v(2) = -1.536848 < 0$$

Because $v(2)$ and $a(2) = v'(2)$ have the same sign, the speed of the particle is increasing at time $t = 2$.



0	1
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The response accurately includes the criteria below.

- Answer with reason

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P3** can be earned by explicitly presenting $v(t) = 0$.
- **P3** can also be earned by identifying $t = 0.356$.

Model Solution:

$$v(t) = 0 \Rightarrow t = 0.356432$$

8-2



0	1
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The response accurately includes the criteria below.

- Considers $v(t) = 0$

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, a response does not need to say $v(t)$ changes sign “from positive to negative,” but it cannot say $v(t)$ changes sign “from negative to positive.”
- For **P4**, a reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The particle changes direction at time $t = 0.356$ because $v(t)$ changes sign there (from positive to negative).



0	1
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The response accurately includes the criteria below.

- Answer with reason

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P5** is earned with a definite integral with correct lower and upper limits in either order, with or without the differential.
 - Correct handling of the sign of a reversal in the limits of integration is not assessed in **P5**.
 - Writing an indefinite integral and never presenting a definite integral does not earn **P5**.

Model Solution:

8-2

$$x(0) = x(5) + \int_5^0 v(t)dt = -4 + \int_5^0 v(t)dt$$



0

1

The response accurately includes the criteria below.

- Integral

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P6** is earned for adding the initial condition to a definite or indefinite integral. This addition may be presented explicitly (e.g., $-4 \pm \int_5^0 v(t)dt$) or implied by the combination of a correct integral and answer (e.g., “ $\int_5^0 v(t)dt = 7.351$, so 3.351”). A response of $x(5) \pm \int_5^0 v(t)dt$ alone has not yet earned **P6**, but it will earn **P6** if the response also explicitly identifies “ $x(5) = -4$ ” or presents the correct answer.
- If the differential dt is missing:
 - Writing $-4 \pm \int_5^0 v(t)dt$ or $-4 \pm \int_0^5 v(t)$ earns **P6**.
 - Because $\pm \int_5^0 v(t) - 4$ may mean $\pm \int_5^0 v(t)dt - 4$ or $\pm \int_5^0 (v(t) - 4)dt$:
 - An integral of $\int_5^0 v(t) - 4$ or $-\int_0^5 v(t) - 4$ in the presence of the correct answer of 3.351 provides sufficient evidence to infer that the correct placement of the differential was used. Such a response earns **P6**.
 - The equation $\int_0^5 v(t) - 4 = -11.351$ or $-\int_5^0 v(t) - 4 = -11.351$ earns **P6**.
 - An integral of $\int_5^0 v(t) - 4$ or $-\int_0^5 v(t) - 4$ in the presence of a correct but inappropriately rounded answer of 3, 3.3, 3.4 or 3.35 earns **P6**.
 - Writing $\int_5^0 v(t) - 4$ or $-\int_0^5 v(t) - 4$ in the presence of no answer or an incorrect answer does not provide sufficient evidence that the correct placement of the differential was used. Such a response does not earn **P6**.

Model Solution:

8-2

$$x(0) = x(5) + \int_5^0 v(t) dt = -4 + \int_5^0 v(t) dt$$



0	1
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The response accurately includes the criteria below.

- Uses initial condition

Part C Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P7** for the correct answer:

- $-4 - \int_5^0 v(t) dt = 3.351$
- $-4 + \int_0^5 v(t) dt = 3.351$
- $\int_5^0 v(t) dt = 7.351 - 4 = 3.351$
- $\int_5^0 v(t) dt = 3.351$

Model Solution:

The position of the particle at time $t = 0$ is 3.351.



0	1
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The response accurately includes the criteria below.

- Answer

Part D Point 8

8-2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned for any of the following, with or without the differential dt :

- $\int_0^5 |v(t)| dt$
- $\int_0^a v(t) dt - \int_a^5 v(t) dt$ for $0.35 \leq a \leq 0.36$, or equivalent
- $-\int_0^a v(t) dt + \int_a^5 v(t) dt$ for $0.35 \leq a \leq 0.36$ (This will produce a negative answer of -7.493 , which can be resolved to give the correct answer.)

- Alternate solution:

$$\int_0^{0.356432} v(t) dt - \int_{0.356432}^5 v(t) dt$$

Model Solution:

$$\int_0^5 |v(t)| dt$$



0	1
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The response accurately include the criteria below.

- Setup

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The total distance traveled by the particle is 7.493 (or 7.492).

8-2



0

1

The response accurately includes the criteria below.

- Answer